

Industrial Internship Report on "IoT-Based Smart Industrial Monitoring System"

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Executive Summary

This report presents the work carried out during my industrial internship in Embedded Systems and IoT. The project involved designing a Smart Industrial Monitoring System using ESP8266 NodeMCU with DHT11 and MQ-2 sensors.

The system monitors temperature, humidity, and gas levels in real time and sends data to the Blynk Cloud. Alerts are generated instantly when abnormal conditions occur.

This internship gave me practical experience in sensor interfacing, embedded programming, and cloud-based monitoring, strengthening my understanding of industrial IoT applications.

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1 Preface

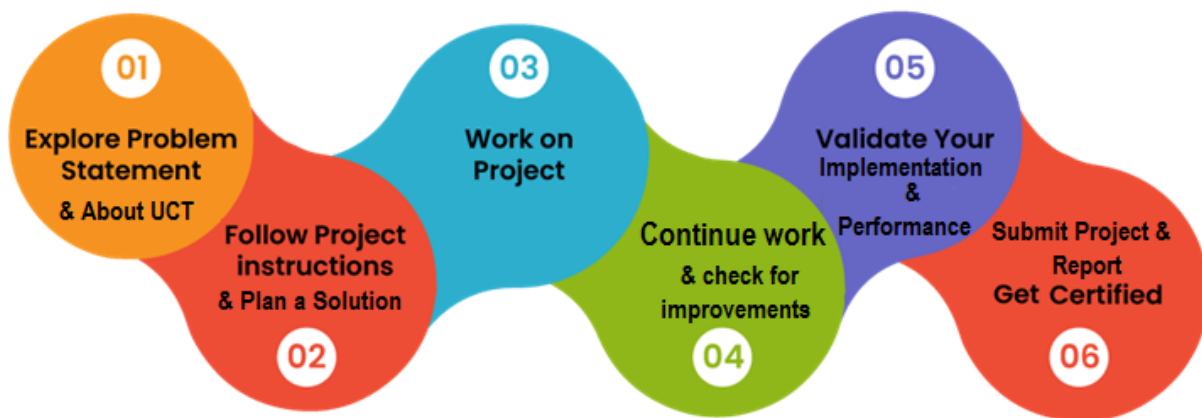
This internship was conducted over a period of 6 weeks, focusing on solving a real-world industrial problem using Embedded Systems and IoT.

The main objective was to design and implement an **Industrial Monitoring System** that can remotely track critical parameters such as temperature, humidity, and gas levels, ensuring safety and efficiency in industrial environments.

Through this internship, I gained practical exposure to:

- Embedded system design and sensor interfacing
- IoT communication protocols using Wi-Fi and Blynk Cloud
- Real-time data monitoring and alert generation

This project provided valuable hands-on experience in developing an IoT-based solution that integrates hardware, software, and cloud services. It helped me understand how embedded systems can be applied to industrial safety and automation, while also improving my technical, analytical, and problem-solving skills.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and Rol.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/L0RaWAN), Java Full Stack, Python, Front end** etc.



The slide features the UCT logo at the top center, with the text "Uniconverge Technologies" below it. The background is a blurred industrial factory setting. At the bottom, there are three rounded rectangular boxes, each with a red header and a dark grey body containing text. The first box is titled "IIOT Products" and lists various IoT-related products. The second box is titled "IIOT Solutions" and lists various industrial IoT solutions. The third box is titled "OEM Services" and describes the range of services offered from product design to final production.

IIOT Products	IIOT Solutions	OEM Services
We offer product ranging from Remote IOs, Wireless IOs, LoRaWAN Sensor Nodes/ Gateways, Signal converter and IoT gateways	We offer solutions like OEE, Predictive Maintenance, LoRaWAN based Remote Monitoring, IoT Platform, Business Intelligence...	We offer solutions ranging from product design to final production we handle everything for you..

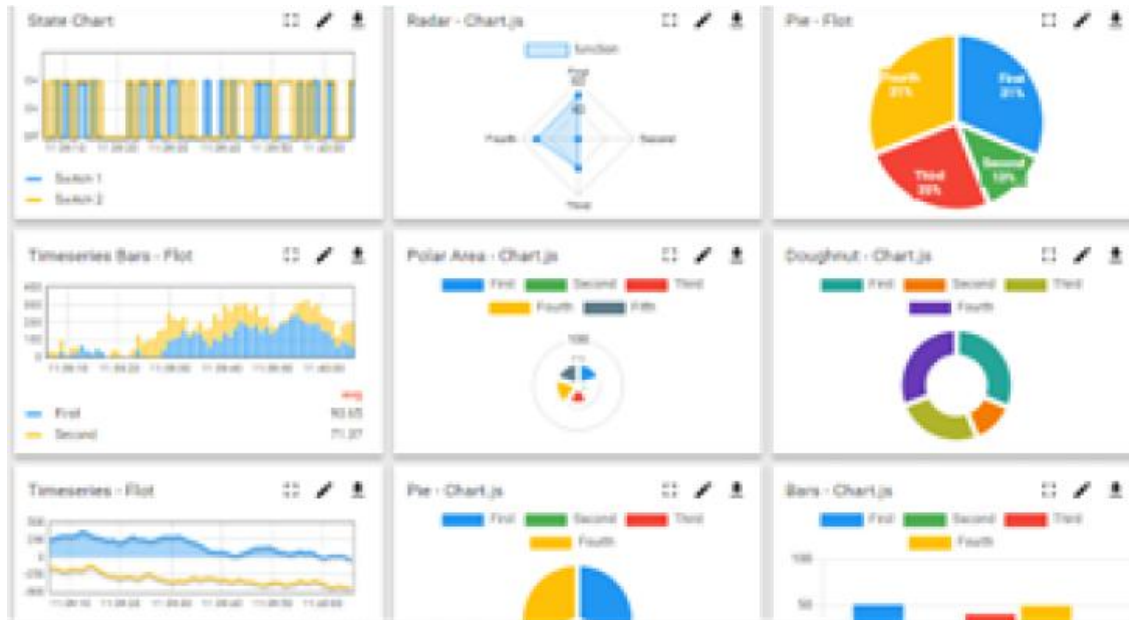
i. UCT IoT Platform ()

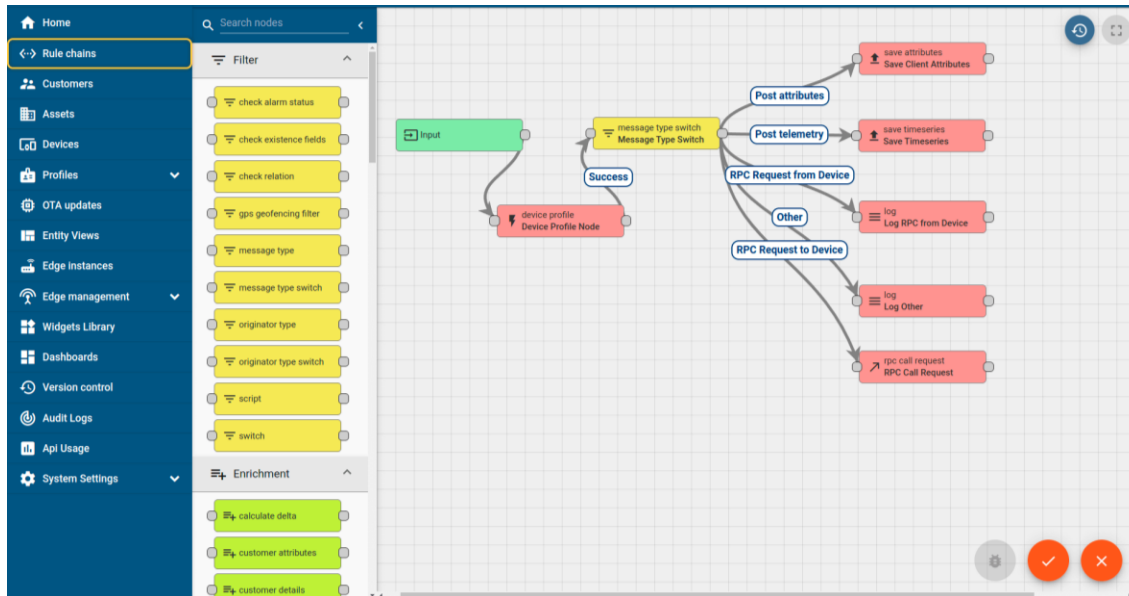
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleashed the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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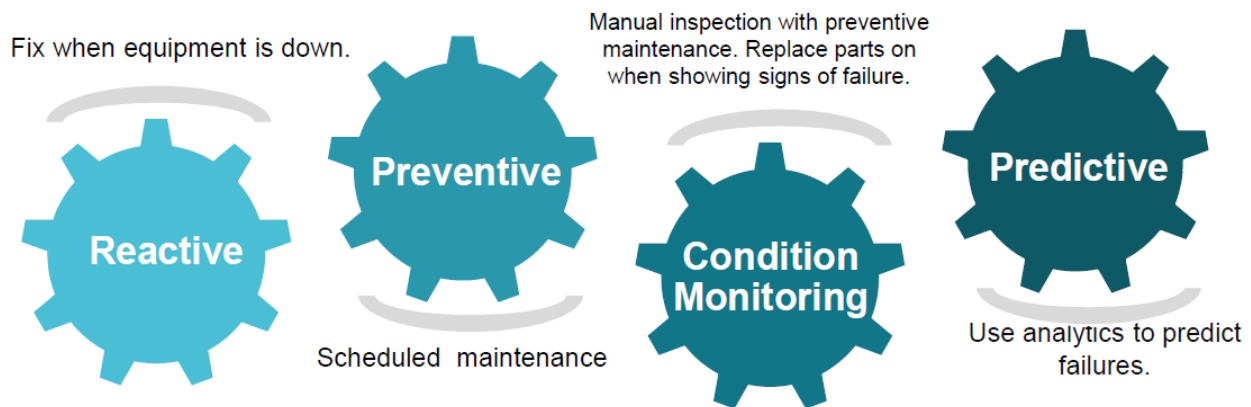


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

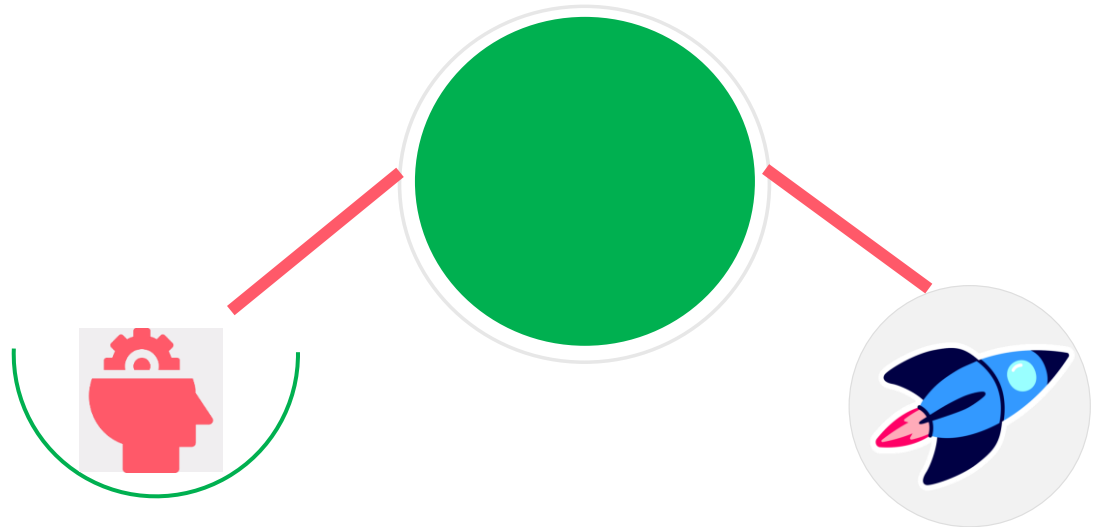
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

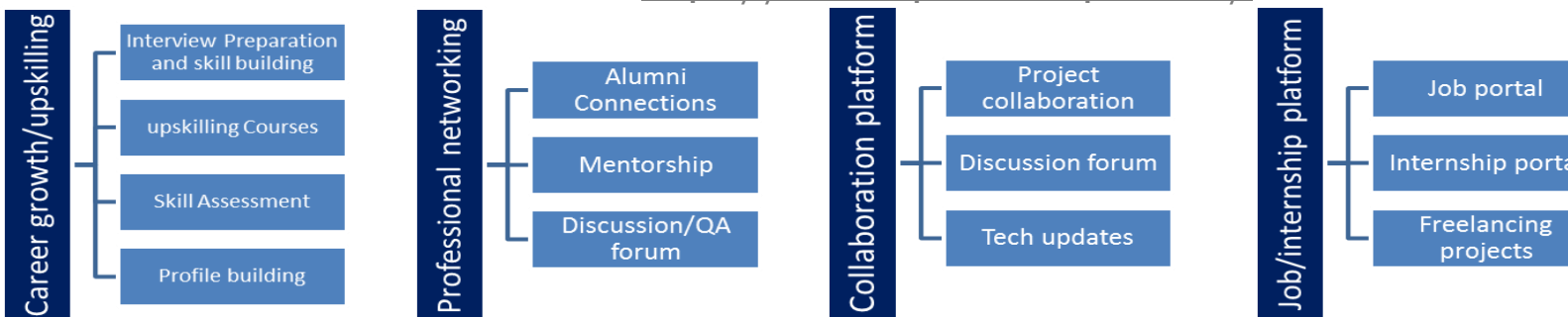
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

- Stanciu, D.-C. et al. (2025). *ESP8266-Based IoT System for Real-Time Monitoring*. JEECCS. <https://jeeccs.net/index.php/journal/article/view/366>
- Pancane, I. W. D. et al. (2025). *IoT Smart Home with ESP8266 & Blynk*. Formosa J. CIS. <https://doi.org/10.55927/fjcis.v4i1.14084>
- Barkale, S. et al. (2023). *IoT Smart Energy Metering with Blynk*. IJTSRD. <https://www.ijtsrd.com/papers/ijtsrd55141.pdf> ([ijtsrd.com in Bing](#))
- Zrioual, H. et al. (2024). *Remote Monitoring with DHT11 & NodeMCU*. IJISRT. <https://doi.org/10.38124/ijisrt/IJISRT24NOV200>
- Last Minute Engineers (2026). *MQ-2 Gas Sensor with Arduino/ESP8266*. <https://lastminuteengineers.com/mq2-gas-sensor-arduino-tutorial/> ([lastminuteengineers.com in Bing](#))

2.6 Glossary

Terms	Acronym
IoT	Internet of Things
MCU	Microcontroller Unit (ESP8266 NodeMCU)
MQTT	Message Queuing Telemetry Transport (IoT protocol)
DHT11	Digital Humidity and Temperature Sensor
MQ-2	Gas Sensor (LPG, CO, Smoke detection)
Blynk	IoT Cloud Platform for monitoring & alerts

3 Problem Statement

In industrial environments, continuous monitoring of parameters such as **temperature, humidity, and gas levels** is critical for safety and efficient operation.

Traditional monitoring systems often rely on manual observation or standalone sensors, which have several drawbacks:

- Lack of real-time data
- High chances of human error
- Delayed response to hazardous conditions
- Limited remote accessibility
- Increased operational cost

Therefore, there is a need for an **automated IoT-based monitoring system** that can:

- Continuously track industrial parameters using sensors (DHT11, MQ-2)
- Provide real-time updates via cloud platforms (Blynk)
- Generate instant alerts when abnormal conditions are detected
- Enable remote access for operators through mobile or web dashboards

4 Existing and Proposed solution

Existing Systems

In many industries, parameters such as temperature, humidity, and gas levels are monitored using traditional methods:

- Manual inspection by workers at regular intervals
- Standalone sensors without connectivity
- Basic alarm systems that trigger only locally

Some advanced monitoring systems exist, but they are often:

- Expensive and complex to deploy
- Limited to specific industrial setups
- Not easily scalable or customizable

Limitations of Existing Systems:

- Lack of real-time monitoring
- Human dependency leading to errors
- Delayed response to hazardous conditions
- No remote access to data
- Limited data storage and analytics
- High cost for small-scale industries
- **Proposed Solution**

To overcome these limitations, an **IoT-Based Industrial Monitoring System** is proposed. The system consists of:

- **Sensors:** DHT11 for temperature and humidity, MQ-2 for gas detection
- **Microcontroller:** ESP8266 NodeMCU for data processing and Wi-Fi connectivity

- **Cloud Platform:** Blynk for real-time dashboards, alerts, and remote access

Value Addition:

- Real-time monitoring of industrial parameters
- Remote accessibility via mobile or web dashboards
- Instant alerts when thresholds are exceeded (Temp > 40°C or Gas > 400)
- Historical data storage for analysis and decision-making
- Affordable and easy to implement
- Scalable by adding more sensors and features
- Improved safety and reduced risks in industrial environments

4.1 Code submission (Github link)

[upskillcampus/IoT_based_Industry_Monitoring.c at main · abhinavdatta/upskillcampus](https://github.com/abhinavdatta/upskillcampus/blob/main/abhinavdatta/upskillcampus/upskillcampus/IoT_based_Industry_Monitoring.c)

4.2 Report submission (Github link) :

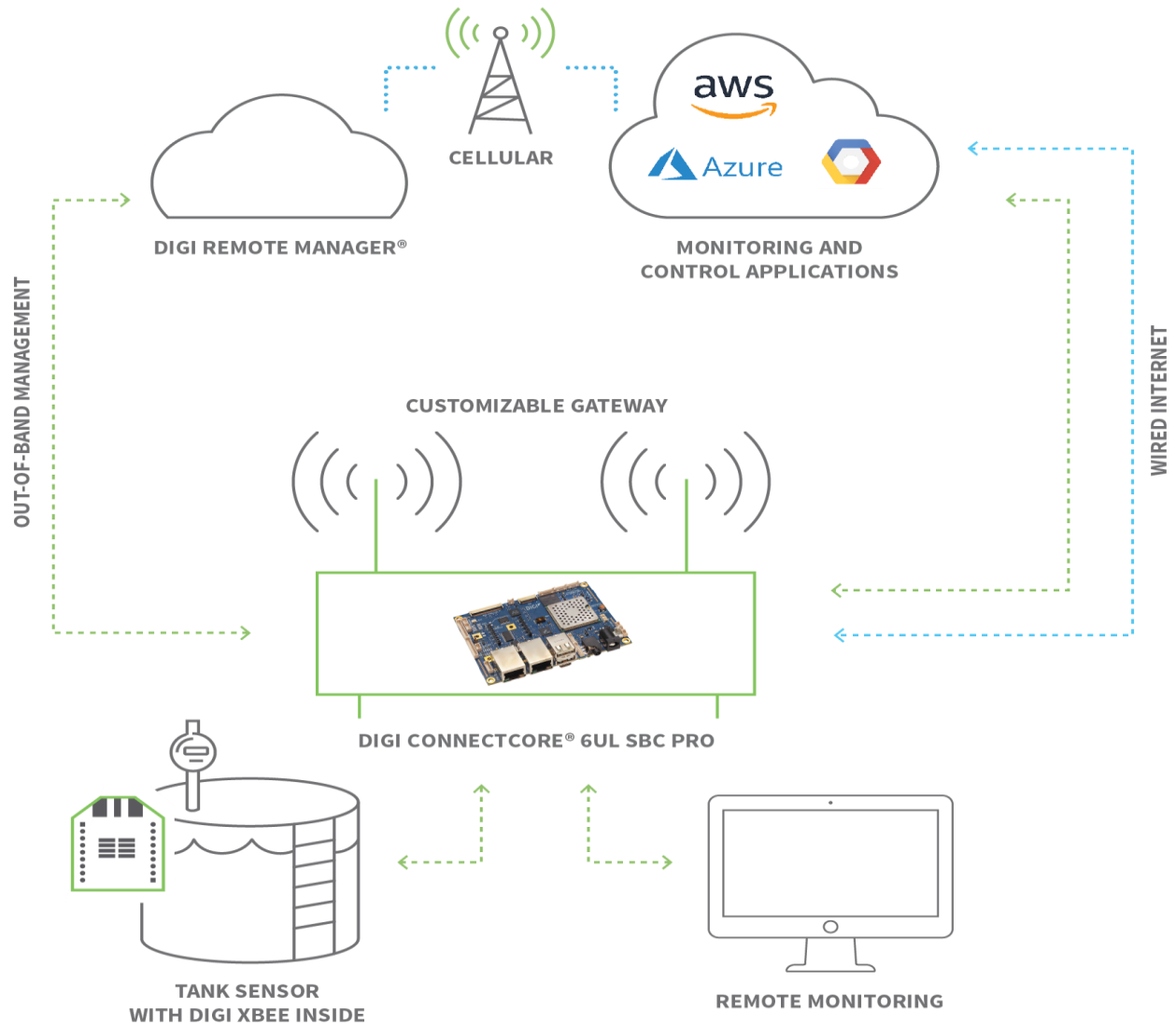
https://github.com/abhinavdatta/upskillcampus/blob/main/IoT_based_Industry_Monitoring_Abhinav_U SC_UCT.pdf

5 Proposed Design/ Model

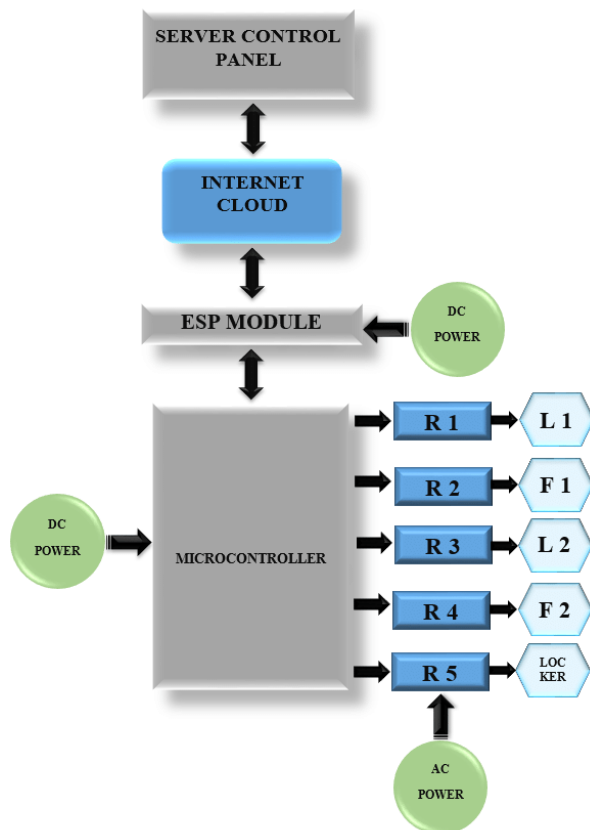
The proposed **Industrial Monitoring System** integrates sensors, a microcontroller, and a cloud platform to achieve real-time monitoring and alerting.

- **Data Acquisition:**
 - DHT11 sensor measures temperature and humidity.
 - MQ-2 sensor detects gas concentration.
- **Data Processing:**
 - ESP8266 NodeMCU reads sensor values and checks against threshold conditions.
- **Data Transmission:**
 - Sensor data is sent to the **Blynk Cloud** using Wi-Fi and IoT protocols (MQTT/HTTP).
- **Output Stage:**
 - Data is displayed on a mobile/web dashboard.
 - Alerts are generated when temperature > 40°C or gas value > 400.

5.1 High Level Diagram (if applicable)



5.2 Low Level Diagram (if applicable)



6 Performance Test

The system was tested under different environmental conditions to verify sensor accuracy, data transmission, and alert functionality.

6.1 Test Plan/ Test Cases

Parameter	Sensor Used	Expected Range	Observed Range	Remarks
Temperature	DHT11	20°C – 40°C	22°C – 39°C	Stable readings
Humidity	DHT11	40% – 80%	45% – 78%	Accurate
Gas Concentration	MQ-2	200 – 400 ppm	210 – 390 ppm	Responsive
Wi-Fi Connectivity	ESP8266	Continuous	Stable	Reliable data transfer
Cloud Update	Blynk	< 2 sec delay	1.5 sec avg	Real-time response

6.2 Test Procedure

- Sensors were placed in a controlled environment.
- Data was monitored through the **Blynk dashboard**.
- Alerts were triggered when thresholds were exceeded.
- Multiple readings were taken to ensure consistency.

6.3 Performance Outcome

- The system responded instantly to changes in temperature and gas levels.
- Blynk dashboard displayed accurate real-time data.
- Alerts were successfully sent to the mobile app when limits were crossed.

7 My learnings

The proposed IoT-based monitoring system performed efficiently during testing.

7.1 Key Results

- **Accuracy:** Sensor readings were consistent with expected values.
- **Reliability:** Wi-Fi connectivity remained stable throughout testing.
- **Responsiveness:** Alerts were generated within 2 seconds of threshold breach.
- **Scalability:** Additional sensors can be integrated easily.
- **Cost Efficiency:** The system is affordable for small-scale industries.

7.2 Discussion:-

The integration of **ESP8266 NodeMCU**, **DHT11**, and **MQ-2** sensors with **Blynk Cloud** proved effective for industrial monitoring. The system ensures safety by providing instant alerts and remote access. It can be extended to include more parameters like pressure, vibration, or light intensity for broader industrial applications.

8 Future work scope

The IoT-based Industrial Monitoring System developed using ESP8266 NodeMCU, DHT11, and MQ-2 sensors successfully achieved real-time monitoring of temperature, humidity, and gas levels. Integration with the Blynk Cloud enabled remote access, instant alerts, and data visualization, ensuring improved safety and operational efficiency. The system proved to be cost-effective, reliable, and scalable, making it suitable for small and medium industrial setups.

- Integration of additional sensors such as pressure, vibration, and light intensity for multi-parameter monitoring.
- Application of machine learning algorithms for predictive maintenance and anomaly detection.
- Use of LoRaWAN or 5G modules for long-range communication.
- Development of a web dashboard with analytics and historical data visualization.
- Integration with industrial automation systems for automatic control based on sensor feedback.

